GPU constructions after Run001: what an A40 certified

Paul Pajo

https://github.com/pageman/ramsey-gpu-constructions

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Abstract

The Kosmos Run001 report set a target of an explicit family with $R(k,k) \geq C^k$ and $C \geq 1.01$. The artefacts from that run are a 1 500-graph feature table, Random Forest and GAT models of Hoffman growth proxies, and four findings about those proxies. They do not include `gnn_model.pt` or a new cell in Radziszowski's survey. This note records a later instrument and a single-A40 campaign (jobs 1a–4c, 28–30 August 2026). The campaign implemented algebraic families named in the Run001 specification, then replaced Hoffman-guided search with Yu's pool, process, and residual decision MIS. No published finite lower bound moved. Paley(17) is still the strongest exact diagonal certificate in the tree ($\omega = \alpha = 3$, $R(4,4) > 17$). Yu's $R(4,20) \geq 252$ is unchanged. Two job-4a rows at $p = 337$ and $p = 353$ were logged as exact $\alpha = 19$ after the C MIS kernel, which stores 256 vertices, returned "not found" on residuals of size 262 and 264 without a timeout flag. Those rows are withdrawn. Job 4c certified $\alpha = 21$ on an 84-vertex K_4-cleaned $W(3,7)$ leftover, so $R(4,22) > 84$, which sits below the published floor 314.

1 Introduction

The Run001 specification asked for a parametric family whose orders grow at least as C^k with $C \geq 1.01$, checked against OEIS A000791, under a budget of about 100 GPU-hours [5]. The prescribed mix was Paley, Frankl–Wilson hybrids, synthetic graphs, and a DIMACS slice. The prescribed learners were a Random Forest and a graph attention network. The prescribed constructor was PPO edge flip. Task 4 asked for a cyclotomic-class search. Task 5 asked for exact clique numbers, SAT, and Lovász ϑ in tiers.

The discovery PDF and the file list from that run stop short of those deliverables. Hoffman bounds stand in for ϑ. There is no shipped PPO policy. A later missing-file note in the same report records that `ramseyconstructions.csv` was absent from the archive at analysis time.

Berghaus and Wagner showed that reinforcement-learning edge flips can lose to random search on $R(4,4)$ [1]. This campaign did not train PPO.

The exponential target was set aside. The specification named GPU-native families that the run omitted. A later criterion is whether one A40 can raise a published finite lower bound in Radziszowski DS1 [4]. The nearby 2026 construction in this size class is Yu's $R(4,20) \geq 252$ on a 251-vertex quintic circulant [8].

Erdős problem 78 stays open; extractor lower bounds of the form $(\log N)^C$ are outside an A40 maximum-clique budget [6]. Problem 986 was settled by container methods [2]. Those results are background. They are not jobs on this pod.

2 The Kosmos baseline

The four Run001 discoveries are kept as statements about surrogates.

Discovery 1 is a compute-capped pipeline: exact ω and α on small N, Hoffman or NetworkX approximations on the rest. That labelling scheme is usable for a training set. It is not a certificate of $R(s,t) \geq n+1$.

Discovery 2 is a sign change in how global eigenvalue features predict a Hoffman growth score C_{Hoff} when the family switches from Paley or cubic residue graphs to general Cayley graphs. That observation supports family-aware ranking. Job 2a later ranked every eligible cyclotomic class-union through $p = 9973$ by Hoffman and produced no survey cell.

Discovery 3 is that degree-preserving edits of Paley graphs worsen $C(\alpha) = \alpha^{1/\alpha}$, which has a unique maximum at $\alpha = e$. Simulated annealing left the Paley basin by raising density and destroying regularity, then matched an Erdős–Rényi cloud at the same density. Job 3b repeated the same score at circulant scale: Hoffman(G) plus Hoffman$(\overline{G})$ never recovered Yu's published connection set S.

Discovery 4 is that Paley had the highest median $C = N^{1/k_{\text{eff}}}$ in the 1 500-graph mix. On exact certificates in this repository the leading diagonal graph is Paley(17). As N grows and k is replaced by a loose spectral upper bound, $N^{1/k}$ falls toward about 1.2. The proxy therefore ranks small exact graphs above large honest spectral rows. That ranking is a property of the KPI.

3 Methods

3.1 Instrument

The code is a Python engine (`python3 -m engine.cli -job ...-scale local|runpod`) and a Next.js catalogue on port 43123. Job ownership lives in `engine/registry.py`. Scale limits live in `engine/scale.py`. Circulant spectra are FFTs of the first row. Boolean Cayley spectra are Walsh–Hadamard transforms. Paley connection sets are images of $x \mapsto x^2$ in $O(p)$ time.

3.2 Survey rule

A graph G on n vertices with $\omega(G) < s$ and $\alpha(G) < t$ implies $R(s,t) \geq n+1$. Spectral k is a ranking key. Plan v2 required a cheap filter in the search loop, a decision $\alpha \leq t-1$ in the certificate, Hoffman out of both, and no dense $(p/2) \times (p/2)$ residual matrix [7].

3.3 Residual reduction

For a circulant, K_4-freeness is equivalent to $G[N(0)]$ being triangle-free. Vertex 0 is non-adjacent to $V \setminus (\{0\} \cup N(0))$, so a residual independent set of size $t-1$ gives $\alpha(G) \geq t$ and rejects $R(4,t)$. An accept requires a proof that no such set exists. Mixed independent sets that meet both $N(0)$ and the residual are not separately bounded in this code. Yu's paper uses the residual; the implementation follows that reduction.

Yu's published distances [8], stored in `data/yu_r4_20.json`, are

$$\begin{aligned} S = \{&1, 2, 4, 9, 10, 13, 18, 25, 26, 33, 36, 37, 43, 45, 50, 52, \\ &65, 66, 71, 72, 74, 79, 86, 90, 93, 100, 103, 104, 107, 109, 119, 121\} \end{aligned}$$

on $p = 251$, $e = 5$, primitive root 6, pool $D_0 \cup D_2$, $|S| = 32$, degree 64, residual 186, $\omega = 3$, $\alpha = 19$. The OpenMP write-up of the missing 19-set on that residual quotes 1.4 seconds and about 2.7×10^7 nodes.

3.4 Decision MIS

`engine/kernels/native_mis.c` is Östergård MIS with $n \leq 256$ and four `uint64` words. Python `bitset_mcs.mis_decision` prefers the C library. A residual independent set of size $t-1$ is a reject. Exhaustion without one is an accept. A timeout is not an accept.

On $n > 256$ the C entry used to return 0 without writing `timed_out`. The ctypes flag stayed 0, so job 4a treated "no 19-set" as exact. The emit path also forced `exact=True`. After the run, $n > 256$ sets a timeout, Python skips `skip_n>256`, and `CELL?` prints only when the decision finished. The regression is `test_mis_n_over_256_is_not_a_certificate`.

3.5 Job 4a

For each prime $p \in [200, 400]$ and each $e \in \{4, 5, 8, 10\}$ with -1 in the index-e subgroup, the job builds every 2-class undirected pool. It runs 64 restricted processes, 64 anneal swaps that reject triangles and greedy $\alpha \geq t$, a multiplier lex-min, and bitset MIS on the eight lowest-greedy survivors. The coded floors are $R(4, 20) \geq 252$, $R(4, 21) \geq 252$, $R(4, 22) \geq 314$.

3.6 Hardware

The pod was one RunPod A40 48 GB (nick `armed_yellow_buzzard`, container `3631e8666026`). Jobs 2a and 4a were almost entirely CPU work billed as GPU time. Run records sit in `data/a40/`. The dashboard still reads the small local `data/catalog.json`.

4 Results

4.1 Local checks

Kernel tests match Paley(17)'s FFT spectrum to `eigvalsh` and give vertex-transitive $\omega = 3$. Yu's S is K_4-free with residual 186. C MIS on Paley(17)'s residual distinguishes $\alpha = 2$ from $\alpha = 3$. An empty residual on 257 vertices is not logged as exact $\alpha < 19$. Local jobs wrote 263 graphs at `scale=local`.

4.2 A40 jobs 1a–3d

Paley recertify (1a) on primes $p \leq 997$ added no new exact diagonal. Gold, Kasami, polarity, Frankl–Wilson, and Sidon rows (1b–1d) are certificates only. The cyclotomic class-union sweep (2a) through $p \leq 9973$ took 99 755 seconds and stored 9 288 Hoffman rows; none is a survey cell. Circulant $R(3, k)$ search (2c) wrote 46 graphs in 29 seconds and stayed behind Coniglio's table for $t \leq 49$ [3]. Block-circulant ILS (3a) added no diagonal cell. Hoffman ILS (3b) wrote about 59 graphs and never recertified Yu's S. Generalized quadrangle scale-up (3c) at $q = 11, 13$ produced Hoffman scores only. ANF search (3d) emitted $n = 13$ ($N = 8192$) and $n = 14$ ($N = 16384$), then stalled: `max_clique` for $n > 64$ is a 64-vertex core plus a Python colouring that has no timeout. Orders $n = 15$ and $n = 16$ were never written.

Job 2a had no checkpoint. A laptop-sleep death at $p = 5347$ forced a restart from $p = 13$.

4.3 Phase 4

Job 4a ran 1 715 seconds and wrote six catalogue rows. The Yu gate passed pool membership, $|S| = 32$, degree 64, residual 186, and a triangle-free $N(0)$. Exact $\alpha = 19$ timed out. The catalogue still marks that row exact from the structural flag. That mark is Yu's published claim, not this kernel.

Job 4b ran 8.56 seconds and wrote zero graphs. Every odd $n \in [501, 521]$ kept residual > 280 after 80 ILS steps from the empty mask.

Job 4c ran 1.58 seconds. The K_4-clean leftover of $W(3, 7)$ has 84 vertices. Greedy α was 20. C MIS found an independent set of size 21 in 1.29 seconds and 1.43×10^7 nodes. Hence $R(4, 22) > 84$. The survey floor used in code is 314.

4.4 Withdrawn 4a rows

`graph_id`	residual	logged	status
`yu_pool_p251_...`	186	$\alpha = 19$ exact	Yu [8]; kernel timed out
`yu_pool_p337_e4`	262	$\alpha = 19$ exact	withdrawn ($n > 256$)
`yu_pool_p353_e8`	264	$\alpha = 19$ exact; `CELL?` 354	withdrawn ($n > 256$)

Four registry lines for $p = 353$ share one `graph_id`. Do not quote $R(4, 20) \geq 338$ or 354. The published lower bound remains 252.

5 Negative statements about this instrument

Ranking whole cyclotomic class-unions by Hoffman for every prime $p \leq 9973$ produced no survey cell. Exact Paley clique numbers in this range are already in Shearer and Exoo. The cutoff 9973 is `cyclo_max`.

Circulant ILS scored by Hoffman(G) plus Hoffman($\overline{G}$) has no gradient toward $\omega \leq 3$ and $\alpha \leq 19$.

The $n > 64$ path in `max_clique` cannot certify a 186-vertex residual. A clique number on a 64-vertex core is a lower bound on ω of a different graph.

A 256-wide bitset MIS that returns "not found" on $n > 256$ without a timeout flag will emit false exact rows if the caller treats that return as a proof. Width belongs in the certificate.

$C = N^{1/k_{\text{eff}}}$ ranks small exact jewels above large spectral rows. It is a diagnostic of the Run001 mix. It is a poor objective for DS1.

GAT training, extra Paley primes, another full class-union sweep, ANF $n = 16$ scored by spectral C, and an AlphaEvolve loop on the pod do not address a residual of width 186–280.

6 Discussion

Run001 Task 4 asked for cyclotomic masks and shipped correlation tables. This repository runs the masks (job 2a) and the 2-class subsets (job 4a). The families in the README gap table have constructors. That closes the omitted-family list. It does not produce $C \geq 1.01$.

Yu's paper is a three-stage pipeline on one prime. The stages are a 50-set pool, a restricted process, and a full branch-and-bound on a 186-vertex residual [8]. Stages one and two are in this tree. Stage three is still the 1.4 second OpenMP run (or Cliquer or SAT). Until that decision is independent and tight, a `CELL?` line is a defect report.

Job 2a's FFTs and job 4a's MIS ran on CPU. Process supervision (tmux versus SIGHUP) consumed more wall time than any GEMM.

Kosmos lost `ramseyconstructions.csv`. This campaign split records across a 14 MB 2a catalogue, a 436 KB 4abc catalogue, an append-only registry, a Mac Downloads tree without `.git`, Cursor Origin (Internal), and public GitHub. The withdrawn 354 bound is stated in the README because a catalogue row with `exact=True` is how a false cell leaves the pod.

7 Claims that are out of scope

Absent from this note: an infinite family with $R(k, k) \geq C^k$ and $C \geq 1.01$; progress on Erdős problem 78; the bounds $R(4, 20) \geq 338$ or 354; an improvement of the published $R(4, 22)$ floor by job 4c; a kernel certificate of Yu's $\alpha = 19$. Hoffman scores, Delsarte bounds, and $N^{1/k}$ are ranking keys.

8 If the criterion is still a survey plus-one

Recertify Yu's residual with a second solver before another hunt. Either widen the bitset past 256 and treat timeout as incomplete, or raise $|S|$ until the residual is at most 256. Prove or drop $\alpha(G) = 1 + \alpha(G[N^c(0)])$ for mixed sets. Put i, j, and a hash of S in `graph_id`. Mark Yu $p = 251$ exact only when `alpha_certified` is true. Seed job 4b from a middle-third sum-free set. Refresh the coded floors from DS1 revision 18 before choosing t at $p = 353$.

If the public negative result is the end of the wave, stop here.

9 Conclusion

Run001 asked for an explicit exponential and delivered a surrogate pipeline in which Paley wins C_{Hoff} and local search leaves Paley only by becoming random-like. This campaign ran the omitted families, spent 28 hours confirming that Hoffman class-unions do not move a survey cell, implemented Yu's search space, and then logged $R(4, 20) \geq 354$ because a 256-bit kernel was read as a 264-vertex proof.

The number of published cells moved is zero. Yu's bound is still 252. The exact diagonal jewel in this tree is still Paley(17). The width that bounded the certificate is 256.

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